Siyuan Feng

BSc Mathematics, University of Warwick | Machine Learning Theory
Email: u5605586@live.warwick.ac.uk | Phone: +44 7303 077764

RESEARCH INTERESTS

Machine Learning Theory: Mathematical foundations of neural-network training, with a focus on learning-rate scaling and transfer across model widths and training budgets, as well as checkpoint averaging.

EDUCATION

University of Warwick Sep 2024 – Jun 2027 (expected)
BSc Mathematics
- Year 1 average: 82.5%; Year 2 average: 82.0% (First-Class level; estimated US GPA: 3.94/4.00).

RESEARCH EXPERIENCE

Learning-Rate Transfer and Scaling under μP Jul 2026 – Present
Research with Prof. Fanghui Liu
- Investigating the mathematical foundations of learning-rate transfer across network widths and training budgets, focusing on optimisation dynamics, scaling behaviour, and maximal-update parametrisation (μP).

Checkpoint Averaging and Learning-Rate Decay Oct 2026 – Present
Research with Prof. Ranko Lazić
- Studying the theoretical relationship between checkpoint averaging and learning-rate decay, particularly how accurately sparse training checkpoints can approximate endpoints of actual decay schedules.

Two-Step Optimal Learning-Rate Geometry under μP May 2026 – Jul 2026
Warwick URSS | Supervised by Prof. Fanghui Liu
- Analysed the geometry of optimal learning rates in deep linear μP networks, establishing mathematical results on generic uniqueness, exceptional nonuniqueness, and stability under perturbations.

SELECTED ACADEMIC PROJECTS

Quantum Computing and Variational Algorithms Jan 2026
- Implemented quantum algorithms using MindQuantum, including quantum Fourier transform (QFT), QAOA for Max-Cut, and VQE for molecular energy estimation; conducted numerical simulations of variational optimisation methods.

Mathematical Computing and Machine Learning Jan 2025 – Mar 2025
University of Warwick – Mathematics by Computer | Supervised by Prof. Damiano Testa
- Implemented and evaluated regression models and neural networks using Python and NumPy, investigating gradient-based optimisation, regularisation, convergence, and generalisation through numerical experiments.

INTERNSHIP & OUTREACH EXPERIENCE

University of Warwick – Undergraduate Mathematics Supervisor Oct 2026 – Present
- Employed by the University of Warwick to lead first-year mathematics tutorials, prepare materials, mark assignments, and provide individual feedback.

Tianma Microelectronics Co., Ltd. – Algorithm Engineering Intern Aug 2025 – Sep 2025
- Built a pixel-level E-paper video ghosting simulation pipeline in MATLAB and Python, covering frame extraction, drive mapping, and grayscale output.
- Replaced pixel-wise loops with NumPy vectorization in a history-frame-difference algorithm, reducing single-run simulation time by approximately 85%.

A-Level Mathematics & Cambridge STEP Tutor Apr 2024 – Aug 2024
- Delivered 100+ hours of mathematics tutoring; supported students who later received offers from Oxford, Imperial College London, NUS, and other universities.

Mathematics Content Creator, Bilibili Ongoing
- Created 60+ original videos explaining calculus and advanced integration problems, reaching 12,000+ followers and 500,000+ cumulative views. Developed accessible problem-solving explanations for a broad online mathematics audience.

AWARDS & RESEARCH FUNDING

University of Warwick URSS Research Bursary — £500 2026
Awarded in support of undergraduate mathematical research.

2025 Quant Championship — 4th out of 200 participants 2025
Competition covering statistics, computational problem-solving, and numerical analysis; awarded a £75 prize.

TECHNICAL SKILLS

- **Scientific computing:** Python (PyTorch: neural-network training; NumPy/SciPy: simulations and learning-rate sweeps; pandas/Matplotlib: data analysis and plots); MATLAB: algorithm simulation.
- **High-performance computing:** Sulis HPC, Linux/SSH, Slurm (GPU job submission and monitoring), PyTorch GPU experiments.
- **Research workflow:** Git (version control), Jupyter, Conda (environments), and LaTeX (mathematical papers).